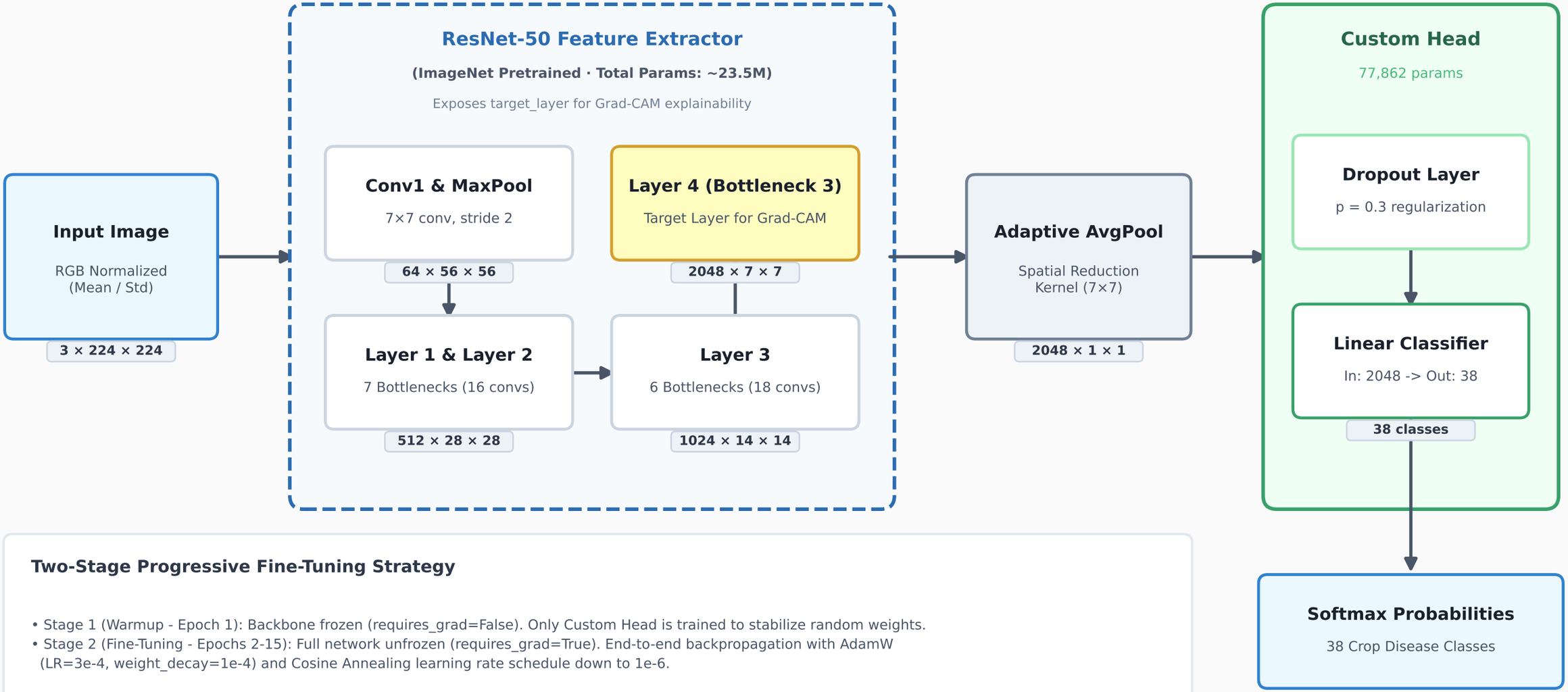
PakAgriVision Classifier Architecture (ResNet-50)

Deliverable 2: Model Architecture, Custom Head, and Two-Stage Progressive Training



Two-Stage Progressive Fine-Tuning Strategy

- Stage 1 (Warmup - Epoch 1): Backbone frozen (requires_grad=False). Only Custom Head is trained to stabilize random weights.
- Stage 2 (Fine-Tuning - Epochs 2-15): Full network unfrozen (requires_grad=True). End-to-end backpropagation with AdamW (LR=3e-4, weight_decay=1e-4) and Cosine Annealing learning rate schedule down to 1e-6.